Daniel Ng

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Education

Purdue University — West Lafayette, IN
Bachelor of Science in Computer Science and Data Science, Graphics Concentration — *Aug 2024 – May 2028*

- Cumulative GPA: 3.83/4.0
- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Systems Programming, Numerical Methods, Linear Algebra, Fundamentals of Computer Graphics, Multivariable Calculus

Experience

Purdue Envision Center – Rosen Center for Advanced Computing — West Lafayette, IN
Simulation Programmer — *Mar 2026 – Present*

- Developed 6+ core features for an interactive Unity-based fluid dynamics laboratory exploring high-resolution scientific datasets
- Eliminated 100% of runtime crashes caused by unbounded heap growth by implementing fixed-buffer memory allocation in C#, maintaining stable memory usage during repeated loading of 100+ MB datasets
- Resolved networking issues and improved multiplayer system stability for CollabXR, an open-source AR/VR classroom built with Unity and the Photon multiplayer SDK
- Reverse-engineered and refactored legacy asynchronous pipelines spanning AWS S3 and Unity runtime systems to restore reliable asset loading

Purdue Data Mine Corporate Partners Program (Caterpillar Inc.) — West Lafayette, IN
Undergraduate Research Software Engineer — *Aug 2025 – May 2026*

- Led a 5 member backend subteam building REST APIs for an AI-powered machine data analysis platform with Python, FastAPI, AWS Bedrock, and Docker
- Architected and implemented an MCP server exposing 5+ custom tools for statistical summaries, histograms, time-series graphs, and geospatial mapping to support data analysis and visualization
- Implemented an LLM orchestration pipeline to automate multi-step analysis and visualization from natural-language queries
- Led weekly demos and coordinated development with Caterpillar staff across biweekly Agile sprints

Fallingdani Games — Albertson, NY
Independent Game Developer — *Jul 2023 – Present*

- Developed 12 Unity/C# games both independently and with teams, working across programming, design, art, and audio
- Shipped multiple releases under strict 2–3 day competition deadlines by rapidly prioritizing features and managing scope
- Placed top 350 out of 9,500+ entries at GMTK Game Jam 2025 through developing novel mechanics for a twin-stick shooter
- Reached 1,700+ views and 700+ plays across 12 total released projects on itch.io

Projects

Falling Engine – Data-Oriented Game Engine | *C++, OpenGL, GLSL, GLFW, CMake* — 2026

- Built a data-oriented C++ game engine with a sparse-set entity-component system (ECS), asset database, and OpenGL renderer, sustaining 60+ FPS with 67,000+ simulated entities and rendered meshes
- Implemented constant-time component lookup using paged sparse storage and dense component arrays in a cache-friendly ECS
- Built a lazy loading pipeline for glTF models with persistent metadata, maintaining stable asset references across imports and reloads
- Decoupled rendering and gameplay systems through a graphics-device abstraction with GPU resource caching for vertex buffers, textures, and shader programs

NYC Service Watch – Data Warehouse and Dashboard | *Python, PostgreSQL, Streamlit, Docker, GitHub Actions* — 2026

- Built an end-to-end ELT pipeline for NYC service request data, implementing incremental ingestion with Polars, PostgreSQL, and dbt
- Improved warehouse reliability through automated dbt data quality tests, layered models, and pre-aggregated analytics
- Developed a Streamlit dashboard with sample data and visualizations to validate ingestion and transformed warehouse outputs
- Automated CI with GitHub Actions and Docker to provision databases, run validation tests, and verify dashboard startup

Echoes of Isovios – 2D Roguelike Platformer | *Unity Engine, C#, Git* — 2025

- Developed 85% of the enemy roster, including 17 enemy classes and all 4 boss battles, for a Steam-published game using Unity, C#, finite state machines and animation events
- Refactored enemy architecture to separate shared logic from specialized behaviors, enabling faster prototyping and balancing
- Collaborated with a 20+ member multidisciplinary team using Git/GitHub to design, develop and ship a complete game

Technical Skills

Languages: Python, C++, C, C#, SQL, Java, GLSL/HLSL, R, HTML/CSS
Frameworks/Libraries: FastAPI, Streamlit, NumPy, PyTorch, Scikit-learn OpenGL, GLM, GLFW
Tools/Platforms: Git, GitHub, Docker, PostgreSQL, GitHub Actions, AWS, CMake, Unity, Photon, Ollama